

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|-------------------------------------|---|
| 1. Slope G | (A) The turning point of a V-shaped (absolute value) graph, where it changes direction. |
| 2. Zero of a function C | (B) A number r between -1 and 1 measuring the direction and strength of a linear association. |
| 3. Rate of change D | (C) An x -value where $f(x) = 0$; the graph's x -intercept. |
| 4. Vertex A | (D) The constant ratio $\frac{\Delta y}{\Delta x}$ — how fast y changes per unit of x . |
| 5. Axis of symmetry H | (E) Predicting <i>outside</i> the range of the data used to build the model. |
| 6. Piecewise function F | (F) A function defined by different rules on different intervals of its domain. |
| 7. Correlation coefficient B | (G) The steepness of a line, $\frac{\text{rise}}{\text{run}}$. |
| 8. Extrapolation E | (H) The vertical line $x = h$ that splits an absolute value graph into mirror halves. |

Teacher Note

Slope (G) and rate of change (D) are close cousins — accept a brief note that both name $\Delta y / \Delta x$; the distinction here is “steepness of a line” (G, geometric) vs. “how fast y changes per unit x ” (D, contextual). Grade on the intended pairing.

Part B — Multiple Choice (12 pts)

Circle the best answer.

- The slope of the line through $(-2, 1)$ and $(4, -2)$ is (A) 2 (B) $-\frac{1}{2}$ ← (C) -2 (D) $\frac{1}{2}$
- The graph of $y = |x - 3| + 1$ is the parent $y = |x|$ shifted (A) left 3, up 1 (B) **right 3, up 1** ← (C) right 3, down 1 (D) left 3, down 1
- $\lfloor -2.4 \rfloor =$ (A) -2 (B) 2 (C) -3 ← (D) 3
- A data set has correlation coefficient $r = 0.12$. The linear association is best described as (A) strong positive (B) strong negative (C) **weak/none** ← (D) perfect
- Which equation is written in **standard form**? (A) $y = 3x - 5$ (B) $y - 2 = 3(x - 1)$ (C) $3x - y = 5$ ← (D) $y = |x| + 3$
- The solution of $|x| < 4$ is (A) $x < 4$ (B) $-4 < x < 4$ ← (C) $x < -4$ or $x > 4$ (D) $x > 4$

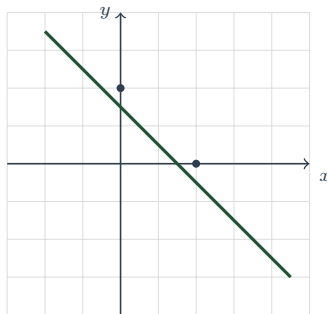
Teacher Note

Item 1: $m = \frac{-2-1}{4-(-2)} = \frac{-3}{6} = -\frac{1}{2}$. Item 3: $[-2.4]$ rounds *down* to -3 (not toward zero). Item 6: “less th**AND**” — $<$ gives a single interval between -4 and 4 ; $>$ would give two rays.

Part C — Short Answer & Computation (40 pts)

Show your work.

1. **Read the graph.** The line f is shown below.



Give the y -intercept: $(0, 2)$; the zero (x -intercept): $(2, 0)$; the slope: -1 . Is f increasing or decreasing? **decreasing**

2. **Write the equation.** Find the equation of the line through $(-1, 4)$ and $(2, -2)$. Write it in slope-intercept form $y = mx + b$. $m = \frac{-2-4}{2-(-1)} = \frac{-6}{3} = -2$. **Using $(2, -2)$:** $-2 = -2(2) + b \Rightarrow b = 2$. **So $y = -2x + 2$.**
Check with $(-1, 4)$: $-2(-1) + 2 = 4$ ✓.

3. **Absolute value transformations.** For $g(x) = -|x - 2| + 5$, state the vertex, the axis of symmetry, whether the graph opens up or down, and the maximum or minimum value. **Vertex $(2, 5)$; axis of symmetry $x = 2$; since $a = -1 < 0$ it opens down; so it has a maximum value of 5.**

4. **Solve.** Solve the absolute value equation. Show both cases.

$$|2x - 5| = 9$$

Case 1: $2x - 5 = 9 \Rightarrow 2x = 14 \Rightarrow x = 7$. **Case 2:** $2x - 5 = -9 \Rightarrow 2x = -4 \Rightarrow x = -2$. **Solutions:** $x = 7$ and $x = -2$. **Check:** $|2(7) - 5| = 9$ ✓, $|2(-2) - 5| = 9$ ✓.

5. **Solve and express three ways.** Solve $|x + 1| \leq 4$. Write the solution as an inequality and in interval notation, then graph it on the number line. $-4 \leq x + 1 \leq 4 \Rightarrow -5 \leq x \leq 3$. **Interval:** $[-5, 3]$.

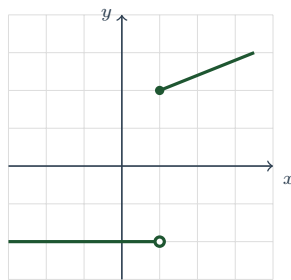


6. **Evaluate a piecewise function.** Let $f(x) = \begin{cases} 2x + 1, & x < 2 \\ 5 - x, & x \geq 2 \end{cases}$

$$f(-1) = -1 \quad f(2) = 3 \quad f(4) = 1$$

Which piece did you use for $f(2)$, and how do you know? **The bottom piece $5 - x$, because the condition is $x \geq 2$ and 2 satisfies $x \geq 2$ (the boundary is included).**

7. **Read a piecewise graph.** The function h is shown below.



$h(-1) = -2$; $h(1) = 2$. Is h continuous at $x = 1$? Explain. **No — there is a jump. The left piece approaches -2 (open dot) while the value at $x = 1$ is 2 (closed dot), so the graph breaks at $x = 1$.**

8. **Linear regression.** A study of x = hours studied and y = test score (for x between 1 and 8 hours) gives the line of best fit

$$\hat{y} = 6.5x + 52.$$

- (a) Interpret the slope 6.5 in context. **Each additional hour of studying is associated with about 6.5 more points on the test.**
- (b) Predict the score for a student who studies 4 hours: $\hat{y} = 6.5(4) + 52 = 78$.
- (c) Is predicting the score at $x = 20$ hours interpolation or extrapolation? Why might that prediction be unreliable? **Extrapolation — 20 is outside the data range (1–8 hours). The linear pattern may not hold that far out (a score can't exceed 100), so the prediction is unreliable.**

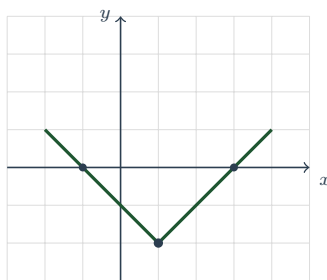
Teacher Note

Item 5: accept “ $-5 \leq x \leq 3$ ” or “ $[-5, 3]$ ” with closed dots on the number line (the $\leq$ makes both endpoints included). Item 8b: full credit for 78 with the substitution shown. Common error on item 6: using the top piece for $f(2)$ because $2x + 1$ “looks” first — reinforce that $x = 2$ fails $x < 2$.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. The absolute value function $g(x) = |x - 1| - 2$ is graphed below.



- (a) State the two zeros of g . (b) On what interval is $g(x) < 0$ (the function negative)? On what set is $g(x) > 0$? (c) Using the zeros, explain *why* the sign of g switches where it does. **(a) The zeros are $x = -1$ and $x = 3$ (where the V crosses the x -axis). (b) $g(x) < 0$ on the interval $(-1, 3)$, where the graph dips below the axis; $g(x) > 0$ on $x < -1$ or $x > 3$ (i.e. outside $[-1, 3]$). (c) A zero is exactly where the graph touches the x -axis, so it is the boundary between output being below the axis (negative) and above it (positive). Because the graph is continuous, the sign can only change by passing *through* a zero — so it flips at $x = -1$ and again at $x = 3$.**
2. A newspaper reports that in towns with more coffee shops, residents live longer, and the correlation coefficient is $r = 0.86$. (a) Describe the direction and strength of this association. (b) Does this prove that drinking

coffee makes people live longer? Explain the difference between correlation and causation, and give one lurking variable that could explain the pattern. (a) **Positive (both increase together) and fairly strong, since $r = 0.86$ is close to 1.** (b) **No. Correlation means two variables move together; causation means one *makes* the other change. A strong r only shows association, not cause. A lurking variable such as town wealth (wealthier towns have more coffee shops *and* better healthcare) could drive both, so we can't conclude coffee causes longer life.**

Teacher Note

Scoring Part D (6 pts each). D1: 2 pts (a) both zeros $x = -1, 3$; 2 pts (b) negative on $(-1, 3)$, positive outside; 2 pts (c) the “sign changes only by crossing a zero” reasoning. D2: 2 pts (a) positive *and* strong (tie strength to $|r|$ near 1); 4 pts (b) states correlation $\neq$ causation *and* names a plausible lurking variable. Do not give full credit for “no” without a reason.